TOPIC C

Finance without frontiers.

An explainable student finance agent for 6 countries and 30 cross-border directions.



STUDENTBRIDGE AI · DEMONSTRATION SANDBOX · OCTOBER 2026

The problem is not currency. It is complexity.

Students face five connected decisions without a single safe workflow.

01

Tuition deadlines

02

Currency spreads and fees

03

Route eligibility

04

Living-cost uncertainty

05

Building local credit

One financial journey.
Many hidden constraints.

The six-country corridor network

Any country can be a sending country OR a study destination; we do not assume every bank route exists.

CHINA

CNY · SEND OR RECEIVE

UNITED KINGDOM

GBP · SEND OR RECEIVE

JAPAN

JPY · SEND OR RECEIVE

SOUTH KOREA

KRW · SEND OR RECEIVE

AUSTRALIA

AUD · SEND OR RECEIVE

BANGLADESH

BDT · SEND OR RECEIVE

6 × 5 = 30 ORDERED PAIRS

An end-to-end financial agent workflow

Task completion is a competition priority (40%).

1

Set your journey

Origin, destination and programme

2

Read the invoice

Exact amount, deadline, funding

3

Check the data

Reference FX, user quotes, source links

4

Apply the rules

Cost, deadline and reserve constraints

5

Explain every choice

Decision Passport and Digital Twin

6

Approve the sandbox

Simulated receipt and audit event

Our innovation: Three connected decisions

These features work together, not as three separate chatbot answers.

01

EduRoute Intelligence

Eligibility, available funds, protected reserve and tuition deadline are hard constraints. Rank only eligible options.

02

Financial Digital Twin

Change FX, unexpected fees or living expenses. See the new budget and whether your reserve survives.

03

Decision Passport

Show every calculation, source, rejected alternative, warning and a plan integrity digest.

Innovation = auditable financial decisions + explicit uncertainty, not unexplained AI output.

Financial logic that can be tested

The AI does not invent a bank rate or change the decision engine arithmetic.

DETERMINISTIC COST MODEL

Required tuition = Invoice − confirmed scholarship

Total debit = Tuition × quoted rate + applicable fees

Then check: eligibility, deadline, funds after reserve and quote expiry.

CRITICAL REFUSAL CASE

No verified or entered suitable route?

The system explains the gap. It does not fabricate a recommendation.

Real facts. Transparent gaps.

Source provenance is a product feature, not an afterthought.

LIVE REFERENCE FX	Frankfurter v2; dated, reference-only, not a transfer quote.
PUBLISHED UNIVERSITY FEES	Official 2026 programme schedules where amount is directly supported.
STUDENT PERSONAL FINANCE	User-entered, self-reported, not independently verified.
PROVIDER TRANSFER QUOTES	User-entered and explicitly unverified without provider integration.
EXECUTION EVIDENCE	MySQL sandbox log and atomic audit chain; not bank settlement.

Security is part of the workflow

We prioritize constraints and limitations over unsupported compliance claims.

Permission tiers

Read data; plan; explicit sandbox approver; local operator.

Human approval

Approval confirmation is required for every sandbox execution.

Safer inputs

Pydantic validation; PDF size/type limits; user quote provenance.

Audit proof

HMAC-chained events, SHA-256 passports, idempotent test payments.

NOT production-ready: no KYC/AML certification, user authentication, or live bank transaction execution.

MySQL-powered demonstration and test evidence

Automated tests, two example corridors, and an auditable simulated execution.

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TESTS PASSING

Deadline, fee, eligibility, expiry, idempotency, API validation.

02

SANDBOX CASES

China → Australia and Japan → United Kingdom.

0

REAL PAYMENTS

Only explicit, local, approved test ledger entries.

MySQL integration smoke test and real screen-recorded demonstration still required.

A responsible foundation for a scalable product

Next: authenticated bank quote APIs, verified corridor compliance, and secure user accounts.

Students deserve financial confidence, not another black-box recommendation.

STUDENTBRIDGE AI / EXPLAIN THE COST. PROTECT THE STUDENT. SHOW THE EVIDENCE.

SOURCE CODE • TECHNICAL REPORT • SECURITY SELF-ASSESSMENT • SANDBOX LOGS